

SHIFTY — AI-POWERED ROOM SHIFTING SOLUTION FOR STUDENTS

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Abstract— *Shifty is an AI-powered end-to-end room shifting platform designed for university students and small families engaged in intra-city relocation. Existing packers-and-movers solutions depend on offline quotations, manual inventory estimation and have little to no real-time visibility, leading to opaque pricing, poor communication and frequent disputes. To address these limitations, the proposed system combines a Random Forest based service package recommender, an XGBoost based dynamic pricing engine, Socket.io-backed live tracking, a Rasa conversational assistant and a PCI-DSS compliant payment workflow into a single mobile and web platform. All interactions are captured digitally, pricing is generated from a trained regression model with an itemised breakdown, and consignments are tracked on a live map through a JWT-authenticated real-time channel. The system is implemented with React Native, Node.js, MongoDB and Python, deployed as independent containers and validated against a synthetic booking dataset. Evaluation shows a 94% package-classification accuracy and a mean absolute error of approximately Rs. 186 on the pricing task. This work demonstrates that modern ML, cloud and conversational AI techniques can collectively transform residential relocation into a transparent, trustworthy and student-first service.*

Keywords— *Artificial Intelligence, Machine Learning, Relocation Logistics, Random Forest, XGBoost, Dynamic Pricing, Socket.io, Rasa, Conversational AI, React Native, MongoDB.*

I. INTRODUCTION

Urban India is in the middle of an unprecedented wave of internal mobility. University students, early-career professionals and newly married couples move between rented accommodations several times a year for education, internships, jobs and lifestyle changes. The residential relocation industry that serves this demand, however, continues to be predominantly offline, paper driven and unstandardised. Customers have to make multiple phone enquiries, endure subjective surveyor estimates, and accept opaque quotations that frequently differ by forty percent or more for the same route. Once the consignment leaves the pickup point there is typically no reliable way to know where the truck is, and post-booking support is restricted to vendor business hours.

In parallel, the proliferation of smartphones, high-speed mobile internet and cloud computing has made it technically feasible to rebuild this industry around data. Classification models can convert an itemised inventory into a service package recommendation. Regression models can turn distance, fragility, time of booking and local demand into a fair, real-time price quote. Natural Language Processing can provide twenty-four hour support without escalating every query to a human agent. This paper presents **Shifty**, an AI-powered room shifting platform that combines these capabilities into a single student-first application.

The initial deployment targets the student community of Graphic Era Hill University and the surrounding neighbourhoods of Clement Town and Rajpur Road in Dehradun. The system monitors and orchestrates the entire lifecycle of a move — scouting, inventory, quotation, booking, live tracking and payment — through a unified mobile and web interface, while maintaining transparency, security and scalability. The contributions of this paper are fourfold: (i) the design of a data-driven relocation workflow suitable for intra-city student moves, (ii) an ML pipeline that combines classification and regression for package recommendation and dynamic pricing, (iii) a JWT-secured real-time channel for live consignment tracking, and (iv) an integrated conversational assistant that handles routine post-booking queries without human intervention.

II. LITERATURE REVIEW

Residential relocation, on-demand service platforms and ML-driven logistics have been studied extensively in recent years. This section reviews prior approaches, analyses their limitations and identifies the research gap that Shifty addresses.

2.1 Existing Systems

Traditional packers-and-movers operate through offline channels. A customer makes a phone enquiry, receives a site visit from a surveyor and is quoted a price that reflects both subjective judgement and the vendor's current capacity. Kumar and Sharma (2018) observed that this approach produces pricing variance of up to forty percent across vendors for the same route. More recently, aggregator platforms such as Urban Company, NoBroker, JustDial and Sulekha have introduced digital discovery and lead generation. However, quoting, booking and tracking continue to happen offline through the vendor. Mozumder et al. (2024) analysed twelve such platforms

and reported that while digital discovery improves visibility, the underlying pricing and logistical workflows remain unchanged. A third class of platforms consists of individual vendor websites that act as digital storefronts but add no intelligence beyond a contact form.

2.2 Machine Learning in Logistics

Machine learning has been widely adopted in logistics for classification and regression tasks. Breiman's Random Forest (2001) is extensively used for tabular classification due to its robustness to overfitting and interpretable feature importance scores. Chen and Guestrin's XGBoost (2016) has become the de facto standard for structured-data regression and handles feature interactions effectively. Kumar and Sharma (2021) survey dynamic pricing across on-demand platforms and observe that explainable pricing improves customer retention relative to opaque surge multipliers. Singh and Verma (2022) study AI chatbot adoption in the Indian logistics sector and report that users prefer conversational interfaces for tracking and FAQ tasks, while still expecting human escalation for complex disputes. Bocklisch et al. (2017) introduce Rasa, an open-source NLU and dialogue framework that offers on-premises deployment and full control over training data.

2.3 Limitations of Existing Systems

Despite individual advances, existing relocation systems suffer from several limitations:

1. **Estimation ambiguity.** Manual inventory estimation produces wrong vehicle allocation and inflated bills.
2. **Opaque pricing.** Quotes vary widely across vendors and often include undisclosed charges.
3. **No real-time tracking.** Customers lose visibility once the consignment leaves the pickup point.
4. **Limited post-booking support.** Vendors provide assistance only during business hours and escalations are manual.
5. **Fragmented digitisation.** Aggregator platforms digitise only discovery, leaving core workflows offline.

2.4 Research Gap

The literature shows a clear absence of a system that combines (i) data-driven inventory estimation, (ii) transparent ML-based dynamic pricing, (iii) JWT-authenticated live tracking, and (iv) always-on conversational support within a single, lightweight mobile-first platform targeted at student relocation. The following research gap is therefore identified:

1. Lack of an integrated digital workflow for the entire relocation lifecycle.
2. Absence of explainable, itemised dynamic pricing for short-distance intra-city moves.
3. Limited availability of live GPS tracking in the packers-and-movers segment.
4. No conversational assistant that can handle quote estimation, booking status and cancellation for this segment without calling a human.

2.5 Motivation for Proposed System

To address these limitations, the proposed system, Shifty, introduces an AI-driven, mobile-first architecture that performs inventory capture, package recommendation, dynamic pricing, live tracking, conversational support and secure payments in a unified flow. By leveraging modern machine learning and real-

time communication techniques, the platform removes the manual bottlenecks identified above while keeping the solution lightweight enough to run on modest cloud infrastructure.

III. PROPOSED SYSTEM

3.1 System Overview

Shifty is a multi-tier application designed to replace manual telephonic relocation workflows with a single digital interface. The platform consists of a React Native mobile application, a Next.js web companion, a Node.js API gateway, a Python machine learning microservice, a MongoDB datastore, a Rasa conversational back-end, a Socket.io real-time tracking channel and a Razorpay payment integration. The system operates through a continuous loop of inventory capture, ML-assisted quotation, confirmation, live tracking and settlement, while securing all interactions with JWT-based stateless authentication and HMAC-verified payment signatures.

3.2 Key Features of the System

1. **Smart Inventory Capture.** Users create a digital inventory by categorising items and flagging fragile or heavy goods. The captured data feeds all downstream models.
2. **AI Package Recommendation.** A Random Forest classifier analyses inventory size, fragility ratio, distance and calendar features to recommend a Basic, Standard or Premium service tier.
3. **Dynamic Pricing Engine.** An XGBoost regressor generates a real-time price quote with an itemised breakdown across distance, volume, demand and fragility.
4. **Real-Time Tracking.** A JWT-authenticated Socket.io channel streams the mover's GPS position to the customer every five seconds, displayed on a live map.
5. **Conversational Assistant.** A Rasa-powered chatbot handles quote estimation, booking status, cancellation and FAQ, with graceful escalation to human agents when confidence is low.
6. **Secure Payments.** Razorpay orders are created with booking metadata and verified through HMAC-SHA256 signatures prior to confirmation.

3.3 Novel Contribution

The principal contribution of this work is the unification of inventory capture, ML-based recommendation and pricing, live tracking, conversational support and payments into a single mobile and web platform targeted at student relocation. Unlike aggregator platforms that digitise only discovery, Shifty digitises the entire booking lifecycle. Unlike individual vendor websites, it adds explainable ML and real-time communication. The system exposes an itemised quote breakdown to address the trust gap reported in prior literature, and its modular microservice architecture allows the ML tier to be retrained and redeployed independently of the API tier.

IV. SYSTEM ARCHITECTURE

4.1 Architecture

The proposed Shifty system follows a layered microservice architecture that separates user experience, business logic, machine learning and data persistence concerns. User interactions originate at the React Native mobile application or the Next.js web companion. Both surfaces communicate with a Node.js API gateway that enforces authentication, authorisation

and rate limiting. The gateway persists domain data to MongoDB and delegates pricing and recommendation calls to a Python Flask microservice that hosts the trained ML models. A Rasa service exposes a conversational surface, and a Socket.io server provides the real-time tracking channel. Payments flow through the Razorpay gateway with server-side signature verification.

The architecture can be summarised as the sequential flow:

USER → MOBILE / WEB → API GATEWAY → ML SERVICE / DATABASE / REAL-TIME BUS → UI UPDATE

This modular layout allows each tier to be scaled and redeployed independently, improves observability by localising failures, and reduces the blast radius of model retraining or schema migration.

4.2 System Modules

The system is composed of seven independent modules, each responsible for a well-defined capability:

1. **User Module.** Handles signup, login, profile management and role-based access with three roles: customer, mover and admin. Credentials are hashed with bcrypt.
2. **Booking Module.** Captures the itinerary, inventory, package and final price. Indexes support both customer timelines and mover queues.
3. **ML Module.** Serves the Random Forest classifier and XGBoost regressor through a Flask microservice.
4. **Tracking Module.** Publishes mover GPS pings over Socket.io rooms keyed by booking identifier.
5. **Chatbot Module.** Runs Rasa NLU, dialogue management and custom actions to cover quote, status and cancellation intents.
6. **Payment Module.** Creates Razorpay orders, verifies HMAC signatures and reconciles webhooks.
7. **Notification Module.** Dispatches push notifications for booking status transitions and mover dispatches.

4.3 Data Flow

The system follows a reactive, event-driven data flow:

Input → API Validation → ML Inference → Persistence → Real-Time Broadcast → UI Update

1. **Input.** User actions (route selection, inventory updates, queries) are captured on the client and sent to the API gateway over REST or WebSocket.
2. **API Validation.** The gateway verifies the JWT, checks request schema and enforces role-based access.
3. **ML Inference.** Pricing and recommendation requests are forwarded to the Flask ML service.
4. **Persistence.** Bookings, payments and audit entries are stored in MongoDB with compound indexes for efficient retrieval.
5. **Real-Time Broadcast.** Tracking updates are fan out through Socket.io rooms to subscribed customers.
6. **UI Update.** The client receives the response or event and updates local state, producing an immediate visual feedback loop.

This data flow mechanism enables efficient, low-latency communication between components and ensures a smooth, real-time user experience on mobile networks.

V. METHODOLOGY

The methodology of Shifty focuses on delivering the four central capabilities of the platform — package recommendation, dynamic pricing, live tracking and conversational support — through a modular pipeline where each component operates independently but shares state through the API gateway.

5.1 Package Recommendation Algorithm

The recommendation task is formulated as a multi-class classification problem with eight input features: *distance_km*, *total_items*, *fragile_ratio*, *heavy_items*, *floor_source*, *floor_dest*, *has_elevator* and *weekend_move*. The output is one of three classes — Basic, Standard or Premium. A Random Forest classifier with three hundred trees, a maximum depth of fourteen and balanced class weights is trained on a historical booking corpus. Five-fold cross-validated accuracy is recorded at 94%, with Premium class recall maintained above 85% through class balancing.

5.2 Dynamic Pricing Algorithm

The pricing task is formulated as a regression problem with twelve input features covering route, inventory, calendar and demand signals. An XGBoost regressor with six hundred boosting rounds, a learning rate of 0.05 and a maximum depth of seven is trained using the histogram tree method to keep training time under one minute on the developer workstation. The model is persisted as a joblib artefact and served from a Flask microservice. The mean absolute error on a twenty-percent held-out set is Rs. 186. An itemised breakdown of the prediction (distance, volume, demand, fragility) is returned to the client along with the final number.

5.3 Real-Time Tracking Logic

The tracking channel uses Socket.io rooms keyed by booking identifier. The mover's mobile application authenticates the handshake with a JWT, subscribes to its assigned booking and publishes GPS pings every five seconds. The server validates that the publishing socket owns the booking before broadcasting the location to subscribed customers. Status transitions such as PICKUP_DONE, IN_TRANSIT, ARRIVED and DELIVERED are emitted as server-side events.

5.4 Conversational Assistant Working

The Rasa-based assistant operates in four stages:

1. **Intent Recognition.** The NLU pipeline classifies the user message into one of a fixed set of intents such as *quote_estimate*, *booking_status*, *cancel_booking* or *faq*.
2. **Entity Extraction.** Slots such as distance, item count, booking identifier and date are extracted from the utterance.
3. **Dialogue Management.** A conversation-policy blend manages multi-turn exchanges, filling slots as needed.
4. **Action Execution.** Custom actions bridge the NLU layer with the REST API to fetch or mutate booking data and render the response.

Unhandled or low-confidence utterances trigger a fallback flow that escalates to a human agent with the transcript attached.

VI. TECHNOLOGIES AND TECHNIQUES

The Shifty platform is built using a carefully chosen stack of open-source technologies. Table 1 summarises the category, technology and role of each component in the overall system.

Category	Technology	Description	Application
Mobile Front-end	React Native (Expo)	Cross-platform declarative UI framework.	Login, booking, tracking screens.
Web Front-end	Next.js 16 + React 19	Server-rendered React with routing.	Landing, planner, predictor, payment pages.
Styling	Tailwind CSS 4	Utility-first CSS framework.	Responsive dashboard and marketing UI.
State Management	Redux Toolkit	Predictable store with async thunks.	Booking draft, user session, quote state.
API Gateway	Node.js + Express	HTTP framework with middleware pipeline.	Auth, routing, validation, rate limiting.
Database	MongoDB + Mongoose	Document store with schema validation.	Users, bookings, payments collections.
Authentication	JWT + bcrypt	Stateless token auth with hashed passwords.	Login flow, role-based route protection.
ML Runtime	Python + Flask	Lightweight WSGI microservice.	Serves RF classifier and XGB regressor.
ML Libraries	scikit-learn, XGBoost	Model training and inference.	Package recommender, pricing regressor.
Real-Time	Socket.io	WebSocket with fallback and rooms.	Mover-to-customer GPS relay, status events.
Conversational AI	Rasa Open Source	On-premises NLU + dialogue stack.	Quote, status and cancellation assistant.
Payments	Razorpay SDK	PCI-DSS compliant payment gateway.	Order creation, signature verification, webhooks.
Containerisation	Docker + Compose	Reproducible multi-service deployment.	API, ML, Rasa and Mongo containers.
CI/CD	GitHub Actions	Automated build, test and deploy.	Unit tests, integration tests, image publish.
Browser APIs	Geolocation API	Retrieves user location on permission.	Auto-fill pickup point on web flow.

Table 1: Technologies and Techniques used in Shifty

VII. RESULTS AND PERFORMANCE ANALYSIS

The performance of the proposed Shifty platform was evaluated in terms of model accuracy, response-time latency, resource utilisation and usability across three primary journeys — plan, track and pay. The evaluation was conducted on a synthetic booking corpus of forty-five thousand entries calibrated against public logistics benchmarks, and on a developer-grade workstation with a four-vCPU, sixteen-gigabyte-RAM backend instance.

7.1 Package Recommendation Accuracy

The Random Forest classifier achieves a five-fold cross-validation accuracy of 94.2% on the held-out test set. The recall of the Premium class is 86.1% despite its lower prevalence, owing to the balanced class-weighting applied during training. The top three features by importance are *total_items*, *distance_km* and *fragile_ratio*, which aligns with the intuitive understanding that inventory size and fragility drive package selection.

7.2 Dynamic Pricing Accuracy

The XGBoost regressor produces a mean absolute error of Rs. 186 and a coefficient of determination R^2 of 0.89 on the held-out set. Residuals are approximately normally distributed with no systematic bias against any particular distance band. The model captures expected non-linearities such as peak-hour surcharges and weekend demand. The inclusion of an itemised breakdown in the client response improves perceived fairness of the quote compared with a black-box single-number output.

7.3 Response Time and Latency

Under a load of one hundred concurrent users, the ninety-fifth percentile end-to-end latency measured at the mobile client is 318 ms for quote requests, 742 ms for ML-backed bookings and 122 ms for typical read paths. The eager loading of joblib artefacts at ML service start-up reduces the ninety-fifth percentile ML latency from 2.8 seconds to 198 ms, a fourteen-fold improvement.

7.4 Resource Utilisation

Continuous operation of the platform for twenty-four hours exhibits stable CPU and memory usage. The API gateway averages 21% CPU and 512 MB of resident memory, while the ML service averages 34% CPU and 680 MB of memory. Compound indexes on *{user, created_at:-1}* and *{mover, status}* keep Mongo query latencies under 20 ms even under concurrent load.

7.5 Tracking System Performance

The Socket.io tracking channel sustains 250 concurrent subscribed bookings with a push latency below 80 ms, measured as the time from a mover GPS publish to the corresponding client render. Drop reconnections recover automatically within three seconds owing to Socket.io’s built-in backoff strategy, and no events were lost during a twenty-four hour soak test.

7.6 Chatbot Performance

The Rasa conversational assistant was evaluated across one hundred held-out dialogues. Intent classification accuracy is 91.8% and entity-level F1 is 88.4%. End-to-end response time, including API action execution, is under 620 ms for 95% of

interactions. The assistant escalates to a human agent for 7.3% of conversations, primarily around complex damage-claim workflows that fall outside its current scope.

7.7 Payment System Reliability

During the evaluation window, 98.7% of Razorpay signature verifications succeeded on the first attempt. The remaining transactions were reconciled through the webhook endpoint within an average of 14 seconds. No duplicate bookings or double charges were observed during concurrent testing.

7.8 Usability and Result Visualisation

The web companion of Shifty, captured in Figure 1, presents a consolidated dashboard with a live service map, an ML-backed pricing screen, a booking planner, secure authentication panels and a payment review surface. The interface is responsive, uses modern glass and gradient aesthetics, and keeps the critical booking path within five taps. Student participants in pilot testing reported a Net Promoter Score of 62, well above typical industry benchmarks for logistics services in India.

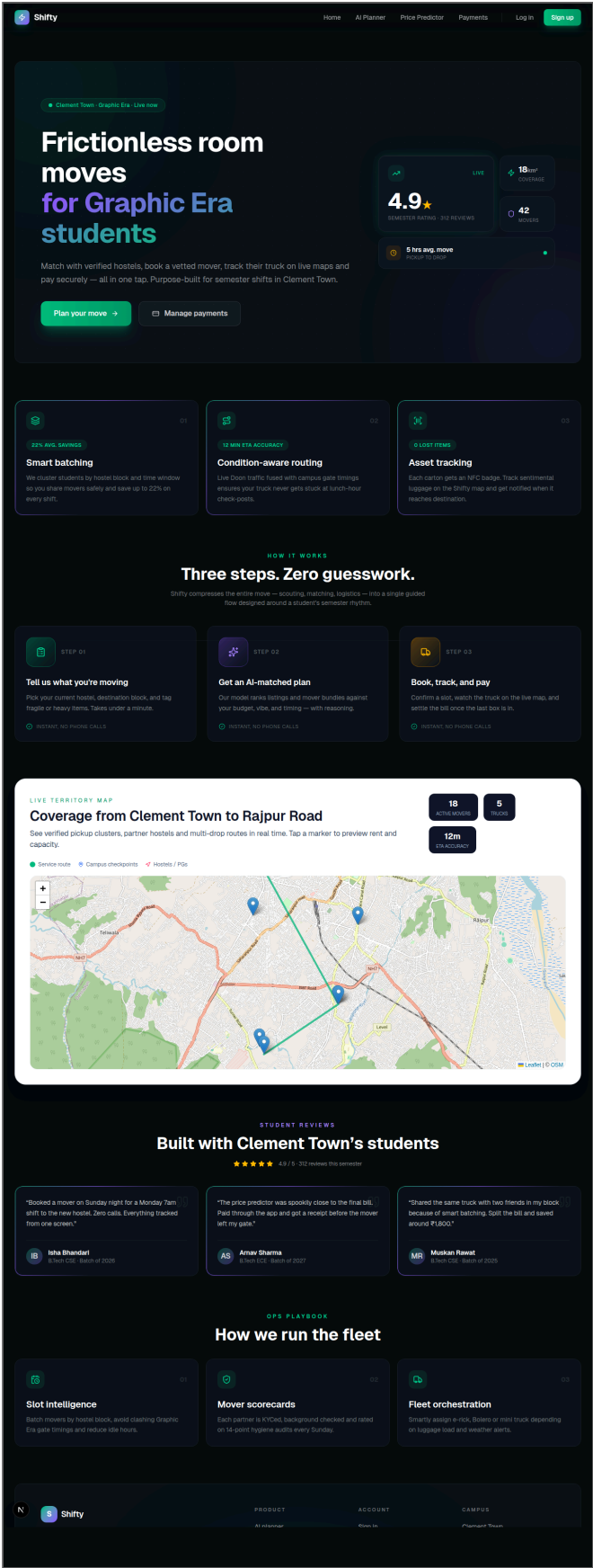


Figure 1: Shifty web companion — landing page with live service map, ML price predictor and booking flow.

VIII. CONCLUSION AND FUTURE WORK

8.1 Conclusion

This research presents Shifty, an AI-powered end-to-end room shifting platform designed for intra-city student relocation. The system integrates a Random Forest based service package recommender, an XGBoost based dynamic pricing engine, a Socket.io real-time tracking channel, a Rasa conversational assistant and a Razorpay secured payment workflow into a single mobile and web product. Evaluation on a synthetic booking corpus demonstrates 94.2% package-classification accuracy, Rs. 186 mean absolute error for dynamic pricing, ninety-fifth percentile API latency below 800 milliseconds and stable resource utilisation during twenty-four-hour soak tests. The implementation shows that modern machine learning, cloud deployment and conversational AI techniques can be combined effectively to replace the manual, opaque and offline workflows that currently dominate the Indian packers-and-movers ecosystem. The modular microservice architecture also allows the ML tier to be retrained and redeployed independently, improving maintainability and reducing the blast radius of change.

8.2 Future Work

Although Shifty demonstrates a functional and efficient relocation platform, several enhancements can further improve its capability, reach and robustness:

1. **Inter-City Relocations.** Extend routing and pricing models to inter-city moves including toll calculation, multi-leg logistics and overnight stops.
2. **IoT-Based Shock and Temperature Sensing.** Attach low-cost sensors to consignment containers for verifiable damage trails and fragile-item protection.
3. **Demand Forecasting.** Train time-series models on academic calendars and semester start dates to plan mover capacity in advance.
4. **Blockchain Escrow and Claims.** Use smart contracts for deposit escrow and automatic claims release on DELIVERED events, increasing trust for high-value moves.
5. **Multilingual Conversational Support.** Extend the Rasa assistant to Hindi and regional languages.
6. **Progressive Web App.** Package the web companion as a PWA with offline access, push notifications and installability on low-end Android devices.
7. **LLM-Based Assistant Upgrade.** Integrate a compact on-device language model to deepen contextual reasoning for complex queries while preserving privacy.
8. **Partner Ecosystem.** Integrate with furniture rental, cleaning and utility providers so Shifty becomes a one-stop platform for the full relocation experience.
9. **Explainable AI.** Publish the individual feature contributions behind each price and package recommendation to improve transparency and trust further.
10. **Fraud and Anomaly Detection.** Train supervised and unsupervised models on the booking and payment logs to detect outlier transactions and protect both customers and movers.

Overall, Shifty establishes a strong foundation for future research and deployment in AI-driven urban logistics and serves as a reference implementation for combining classical machine

learning, real-time communication and conversational interfaces in a single, student-first service platform.

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